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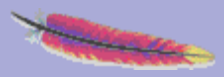
MO18: Using Tomcat to build Desktop Applications

Will Iverson
Cascade Technology Group

ApacheCon US 2003
November 16-19, 2003

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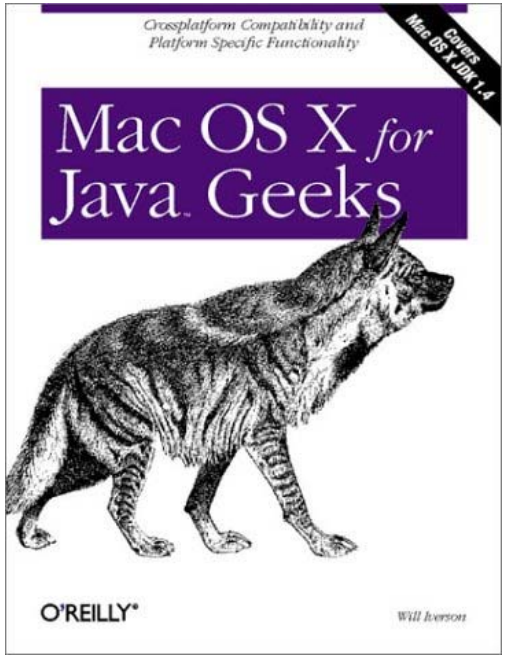
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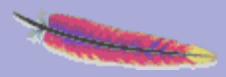
Who Am I?

- Author of the recently published O'Reilly book "Mac OS X for Java Geeks"
- Led Cascade Technology Group, a private consulting firm, since 1999
- Former Developer Relations Manager for Visual Café @ Symantec
- Former Java & Runtimes Product Manager at Apple Computer
- **In brief, a huge tools & platforms geek!**



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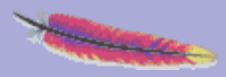
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Summary

- A close look at how a Java-based web application can be converted to a desktop application for use on both Mac OS X and Windows systems.
- Particular emphasis will be paid toward understanding how to build complete packages that are easy for an end-user to install, plus solutions for technical and installation problems.

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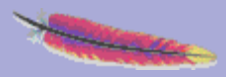
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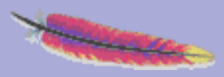
Agenda

- Basic Idea
  - Background
  - Similar Concepts
  - Existing Examples
  - Downsides
- The “Stack”
  - Presentation
  - Web Container
  - Persistence Integration
  - Persistence Engine
  - Development Environment
  - Installer/Packaging



Backstory

- Where did this idea come from?
  - ASP!
  - Server application!
  - Desktop application!
- http://www.cascadetg.com/

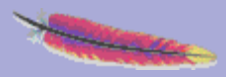


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Basic Idea

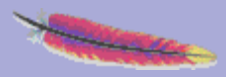
- **Make a web application as easy to install & use as an ordinary desktop application**

- Allow for single desktop, LAN, ASP, customer IT hosted… doesn't matter.
- Best choice for the situation as determined by the *user,* not the developer.



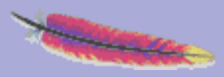
Basic Idea *con't*

- Evolution of client/server
  - Part of a very large (ongoing) debate…
  - What exactly is a server these days?
- Software UI Challenge
  - Locked into a model (native/Swing/web) at absolute earliest stage of development
  - Desktop apps that need to add "multi-user" support…
  - Web apps that are too complicated for small business/local office use…
  - Customers don't want to give up data to an ASP…



Basic Idea *con't*

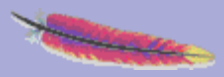
- Keeps your UI presentation *portable*
  - Easier to find good web people than good Swing people
  - Easier to build portable web UI than portable Swing UI
- Keeps your processing logic *portable*
  - Code goes on best system
  - Access via the best (most convenient) system



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Similar Concepts

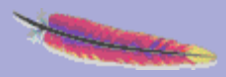
- UI Access
  - VNC, X, “Go To My Desktop”, PC Anywhere…
- Self-Updating Rich GUI
  - Java Web Start
  - Actually, this is very complimentary…
    - E.g. start Web Start as a background service… http://www.seajug.org/vqwiki/jsp/Wiki?JavaWebStartTips
- “True” Digital Hub
  - PC, Mac, Linux Access
  - PDA 802.11b Access
  - Wireless access to home network resources
  - Etc, etc, etc…



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Existing Examples

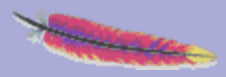
- CUPS
  - Running Mac OS X 10.2+ or some Linux?
  - Try it now - **http://localhost:631/**
- Radio Userland
  - http://www.userland.com/
- WebLogic, JBoss, & other application servers
- WebAdmin
  - http://www.webmin.com/
- ZOË
  - http://guests.evectors.it/zoe/
- Built-in web app in your router/wireless
  - (e.g. Linksys)



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Downsides

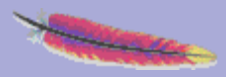
- Terrible idea for certain kinds of application
  - e.g. terrible for a drawing application
  - Not good for "push" UI back to the user
  - "HTML sucks"
- Networking is surprisingly hard
  - The "port problem"
  - Rendezvous is going to be **huge** for solving this!
- "Server" people have a very different mindset from "desktop" people
- ***These problems are all solvable…***



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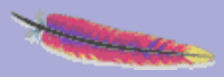
The Stack

- Web Container (Tomcat)
- Presentation
- Persistence Engine (e.g. hsqldb)
- Persistence Integration (e.g. Hibernate)
- Development Environment (ant…)
- Installer/Packaging (e.g. InstallAnywhere)
- *Alternatives easy to imagine*



Web Container

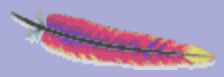
- Using Apache Jakarta Tomcat
  - Pros: Very well known, lots of features, active development, flexible license
  - Cons: Considered “big” by embedded standards
- Alternatives include…
  - Jetty
    - Pros: high speed, small, good license
    - Cons: Not as well known as Tomcat, uses Tomcat JSP engine (Jasper) for JSP development anyways
    - http://www.mortbay.org/jetty/index.html
  - Resin (commercial) http://www.caucho.com/resin/
  - Simple (GPL) http://simpleweb.sourceforge.net/



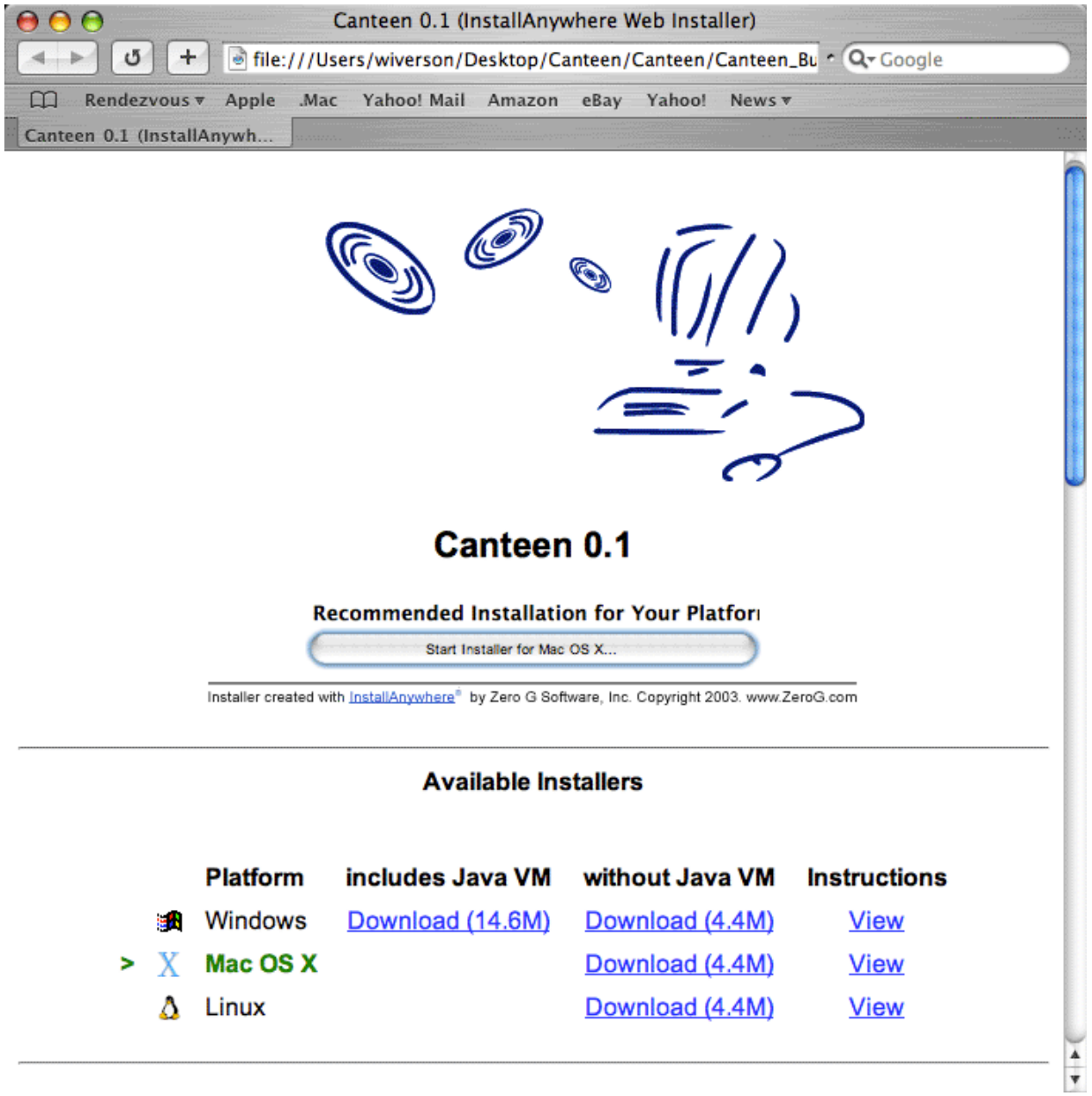
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Different Perspective

- Not just a WAR
- Providing complete server!
- APP_ROOT
  - CONF
  - LOGS
  - TOMCAT_LIBS
  - WEBAPPS
    - ROOT
      - YOUR WAR CONTENTS
  - WORK
- But, this is all “invisible” to the user – they just launch and go…

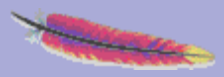


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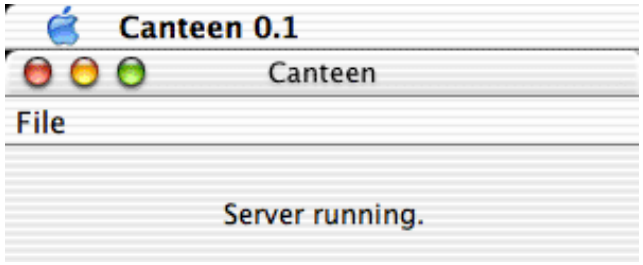
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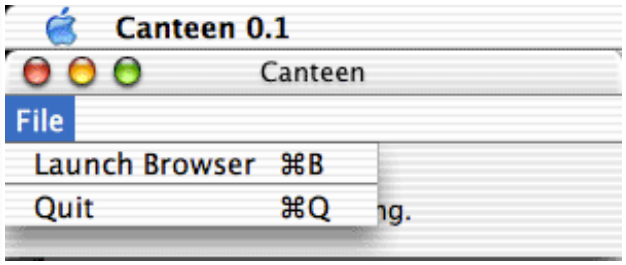
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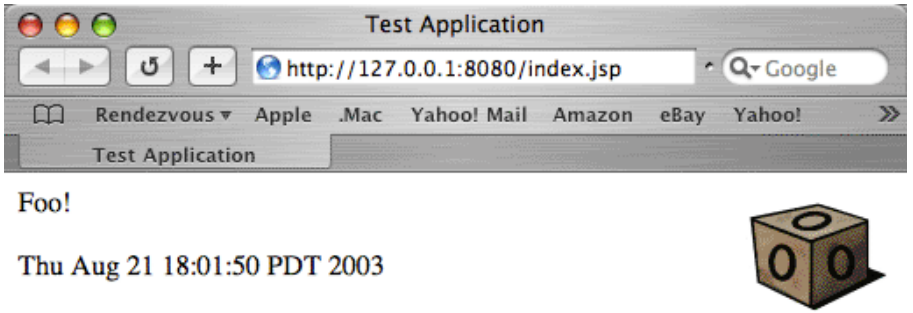


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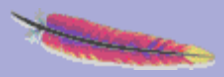






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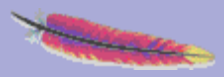
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```
package onjava;
import java.net.*; import java.io.*; import java.awt.*;

public class TomcatWrapper
{
   public static void main(String[] args)
   {
      try
      {
         ServerUI myFrame = new ServerUI();
         myFrame.setStatus("Launching...");
         myFrame.show();

         // This will create a reference to a file in the
         // current working directory, which is the path
         // where the application started (at least on
         // Win32 & Mac OS X)
         File baseDirectory = new File("");
```



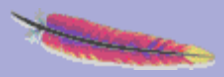

```
            // This is the path to the application's base directory
            String path = baseDirectory.getAbsolutePath();
            String[] temp = {path};

            // Launches the server
            onjava.EmbeddedTomcat.main(temp);

            // Wait a second to be sure the server is ready
            Thread.sleep(1000);

            myFrame.setStatus("Server running.");

            // Launches web browser pointed to the application
            edu.stanford.ejalbert.BrowserLauncher.openURL(
                  "http://127.0.0.1:8080/");
         }
         catch (Exception e)
         {
            e.printStackTrace();
         }
      }
};
```



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```
public void startTomcat() throws Exception
  {
     Engine engine = null;
     // Set the home directory
     System.setProperty("catalina.home", getPath());

     // Create an embedded server
     embedded = new Embedded();

     // Create an engine
     engine = embedded.createEngine();
     engine.setDefaultHost("localhost");

     // Create a default virtual host
     host = embedded.createHost("localhost", getPath()
     + dirchar + "webapps");
     engine.addChild(host);

     // Create the ROOT context
     Context context = embedded.createContext("",
     getPath() + dirchar + "webapps" + dirchar + "ROOT");
     host.addChild(context);

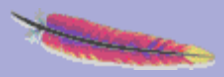
     // Install the assembled container hierarchy
     embedded.addEngine(engine);

     // Assemble and install a default HTTP connector
     Connector connector =
     embedded.createConnector(null, 8080, false);
     embedded.addConnector(connector);

     // Start the embedded server
     embedded.start();
```

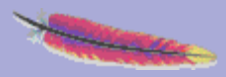
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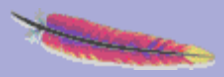
Didn't Manage to Write That Down…

- Article on the O'Reilly OnJava website
- http://www.onjava.com/pub/a/onjava/2003/09/17/macosxjava.html
  - Link on www.cascadetg.com
- Future version could easily use Java Web Start…



Presentation

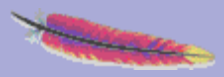
- Easy to build basics
- Strongly recommend template system
- Very pleased with SiteMesh
  - Based on servlet filters
  - Define one or more “decorator” JSP pages
  - Remove all but absolute minimum from JSP pages
  - Add meta tags to application page to control decorator
  - http://www.opensymphony.com/sitemesh/



SiteMesh Page Example

- page.jsp

```
<HTML>
<HEAD>
   <TITLE>Hello World</TITLE>
   <META name="nav" content="main">
</HEAD>
<BODY>
   Hello World!
</BODY>
</HTML>
```

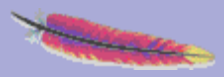


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SiteMesh Decorator Example

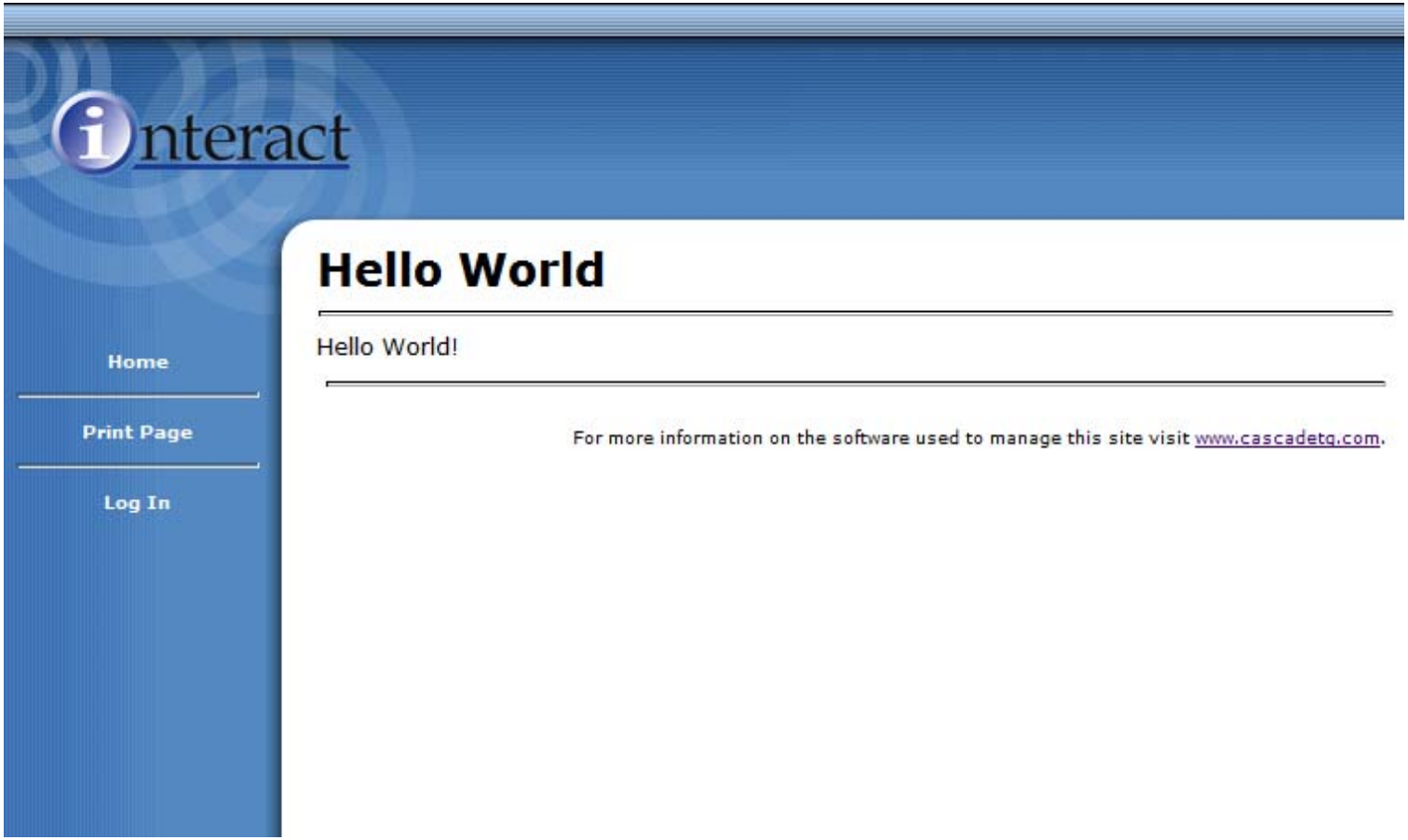
- decorator.jsp

```
<HTML>
<HEAD>
    <TITLE><decorator:title /></TITLE>
</HEAD>
<BODY>
    <H1>Welcome to <decorator:title /></H1>
    <TABLE><TR>
        <TD><!- navbar --></TD>
        <TD><decorator:body /></TD>
    </TR></TABLE>
</BODY>
</HTML>
```



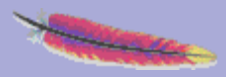
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Interact Example



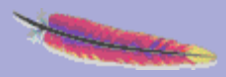
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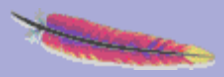
Important UI "Widgets"

- Log in/log out
- Navigation bar
  - Analogous to menu bar in traditional app
- Lengthy task feedback
  - Equivalent to "progress bar" dialog
  - <meta HTTP-EQUIV="refresh" content="60">
- Support different user agents (aka browsers)
  - e.g. Return different page for MSIE/Safari/Camino than for Blazer (the Handspring/Palm browser)
  - List at http://www.psychedelix.com/agents.html



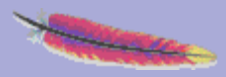
Persistence

- File-based
  - Essentially, same model used by normal desktop apps
  - Requires some sensitivity
    - Implementation to handle sessions
    - Keep file “open” only during single request, etc. or
    - Have time-out/recovery mechanism.



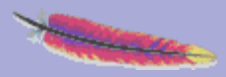
Persistence *con't*

- Relational Database
  - Embedded for small installations
    - Support external for bigger applications
  - SQL for most web applications, natural choice
  - Allows you to expose your data in another format
    - e.g. potentially connect via JDBC to another app
  - Known to be scalable
- Alternative
  - Serialization, custom object db…?



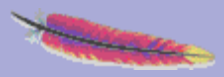
How To Choose…?

- IMHO, Relational Database most powerful, scalable, familiar option
- Several good choices, selected hsqldb
- http://hsqldb.sourceforge.net/
- Pros:
  - Good runtime schema control (ALTER TABLE support)
  - Acceptable management GUI
  - Perhaps most important, Apache-style license



Alternative Choices

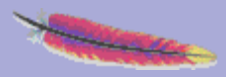
- Mckoi, now 1.0! (GPL)
  - http://www.mckoi.com/database/
- Axion, http://axion.tigris.org/
  - No ALTER TABLE
- Pointbase, Cloudscape, Commercial
- Ozone, LGPL object database
  - http://www.ozone-db.org/frames/home/what.html
- Apache Xindice, XML database
  - http://xml.apache.org/xindice/



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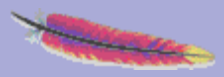
Persistence Integration

- Need to start, stop, clean, etc.
- ALTER TABLE important to be able to update schema
- Currently, using home-grown schema management
  - Hard to support lots of different dbs
  - Extensive evaluation
    - Hibernate
    - Apache DB



Development Environment

- Ant is the core
  - Use the Jasper jspc to precompile JSPs
  - Task to build “bean dev” environment
  - Task to build “jsp dev” environment
  - Everything automated
- CVS to manage source
- Depending on developer, BBEdit, Project Builder, Eclipse, NetBeans, emacs…
  - CVS for all builds, so less critical…
- For front end, mostly Dreamweaver MX
  - Excellent for prototyping



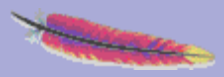
Ant: JSPC Target

```
<target depends="init,jar" description="builds wwwroot jsp
into a java src" name="jspc">
    <delete dir="${jspbuilddir}" />
    <mkdir dir="${jspbuilddir}"/>

<copy todir="${jspbuilddir}" includeEmptyDirs="false">
        <fileset dir="../webapps/ROOT/" />
</copy>
<delete>
    <fileset dir="${jspbuilddir}/decorators/">
        <include name="*.jsp" />
    </fileset>
</delete>

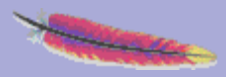
<jspc destdir="${jspbuilddir}" srcdir="${jspbuilddir}"
webinc="wwwroot.xml" />
<echo message="Compiled JSPC JAR" />
</target>
```

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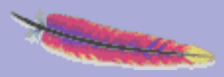
Packaging & Installation

- Java apps (including web servers)
  - Clump of JARs
  - Configuration files
  - Arranged in a set of directories
- But, you need the JVM for non-Mac OS X



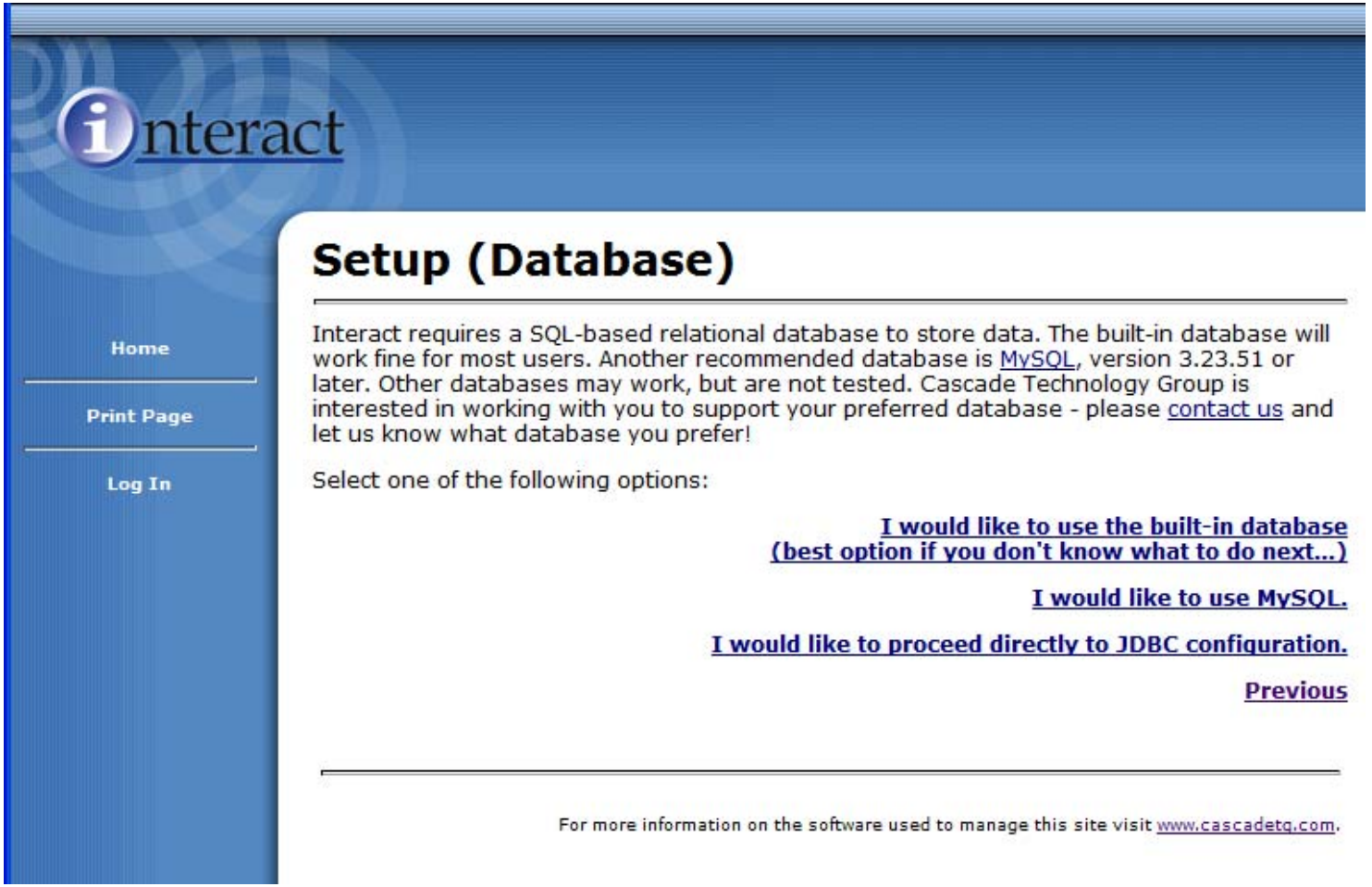
Interact Solution

- Use InstallAnywhere to build the installer
  - Generates Mac OS X, Win32, Linux, Solaris…
- Bundle all of the JARs in a lump for the installer
- Add a very simple GUI for controlling the server
- All other configuration happens in the browser
  - In particular, users walked through db initialization



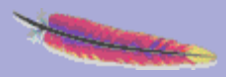
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Configuration Screen



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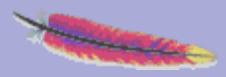
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Tomcat/Canteen Wish List

- Tomcat/Canteen "owning" install
  - Allow single-click WAR installation
  - Pattern off of JNLP/Java Web Start
  - Maybe a new file type?
  - Perhaps a new web application to provide add/remove?
- Reliable JSPC
  - Decreases download
  - Increases startup performance

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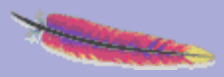


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Not Covered…

- Non-Java environments
- Flash?
- Lots of UI issues
- Synchronization
- Networking
- Security
- Protecting Your IP
- Licensing
- *More – in a sense, a new platform…*

http://www.cascadetg.com/canteen/



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Summary

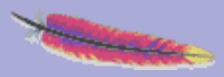
- Basic Idea
  - Background
  - Similar Concepts
  - Existing Examples
  - Downsides
- The "Stack"
  - Presentation
  - Web Container
  - Persistence Integration
  - Persistence Engine
  - Development Environment
  - Installer/Packaging

Another arrow in the quiver

Like regular web dev, w/more packaging & more opportunity

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Demo

– and –

Q & A

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Click anywhere to return.

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